# Analysis Quantifying the Impact of AI Personalization Platforms on Attention Dispersion and Fake News Propagation


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# 1 Introduction

The rapid growth of AI-driven recommendation systems on platforms like TikTok and Instagram Reels has transformed digital content consumption by creating highly personalized, continuous engagement. While enhancing user interaction, these technologies raise concerns regarding reduced attention spans, cognitive distraction, and the accelerated propagation of misinformation [5, 4]. This project utilizes a data analytics approach to explore the interplay between AI personalization platforms, human attention dispersion, and news credibility.

## 1.1 Problem Statement

AI recommendation algorithms maximize engagement through continuous content loops that risk degrading human attention spans. Furthermore, personalized filtering traps users in informational "echo chambers," diminishing their capacity for critical evaluation and increasing vulnerability to fake news propagation [4, 3].

## 1.2 Project Objectives

This project addresses the problem through the following objectives:

- Quantify the relationship between AI recommendation engagement and user attention spans.
- Evaluate how personalized content streams influence a user's capacity to distinguish credible from fake news.
- Extract statistical insights to inform the design of responsible digital consumption behaviors.

# 2 Dataset Overview

## 2.1 Data Sources

To evaluate digital behavior against content credibility, the project integrates three distinct datasets:

1. **AI Recommendation Impact Dataset (ai_recommendation_impact.csv):** Sourced from Kaggle (Social Media Addiction and Mental Health package) [1]; captures user demographic indicators, daily screen time, and digital distraction metrics.

2. **Misinformation Dataset (gossipcop_fake.csv):** Retrieved from the Awesome Fake News Datasets repository [2]. Rooted in the FakeNewsNet framework [3], it contains articles verified as unreliable by GossipCop.

3. **Credible News Dataset (gossipcop_real.csv):** Sourced from the same FakeNewsNet repository [2, 3] to serve as an authentic baseline control group for comparative analysis.

## 2.2 Dataset Size and Scope

- **Behavioral Data:** Contains demographic and behavioral variables to quantify attention levels across short-form video platforms.

- **News Credibility Data:** Comprises textual records and titles. The gossipcop_fake and gossipcop_real corpora are merged programmatically using Python, with a binary target variable (is_fake) engineered for downstream classification.

# 3 Phase 1: Data Exploration & Quality Check

In this stage, a comprehensive structural inspection and quality assessment were performed on the three datasets to understand their overall structure, identify the characteristics of the available variables, and evaluate data quality before proceeding with data merging and advanced statistical analysis.

## 3.1 Dataset Structure and Size

The initial exploratory analysis revealed differences in the size and structure of the three datasets. Table 1 provides an overview of the number of records, variables, and the overall structure of each dataset.

Table 1: Overall Structure and Size of the Datasets

| Dataset Name | Rows (Records) | Columns (Features) | Nature and Types of Variables |
|---|---|---|---|
| gossipcop_real | 16,817 | 4 | Textual and Categorical (IDs, Titles, URLs) |
| gossipcop_fake | 5,323 | 4 | Textual and Categorical (IDs, Titles, URLs) |
| ai_recommendation_impact | 50,000 | 11 | Mixed: Textual/Categorical + Numerical Indices |

## 3.2 Statistical Outputs and Behavioral Indicators

To better understand the distribution and behavioral characteristics of users within the AI recommendation dataset, key descriptive statistical measures were calculated. The resulting insights and statistical indicators are summarized in Table 2.

Table 2: Descriptive Statistical Summary of Digital Behavioral Indicators

| Digital Indicator | Mean | Actual Range | Statistical Interpretation |
|---|---|---|---|
| Echo Chamber & Personalization Index | ~50% | 0 to 100 | Well-balanced sample distribution. |
| Digital Addiction Probability | 49% | 0% to 100% | High density of digital dependency signatures. |

## 3.3 Data Quality Assessment

A data quality check was conducted to identify missing values before proceeding with the analysis. The results are summarized in Table 3.

Table 3: Data Quality Assessment and Missing Values Status

| Digital Indicator | Data Quality Status | Missing Values (NaN) | Action Taken |
|---|---|---|---|
| News Datasets (Real & Fake) | Minor Cleaning Required | Found in URL & Tweet ID | Documented for Clean Phase |
| AI Behavioral Indicators | 100% Clean | None Across All Records | Ready for Merging |

# 4 Phase 2: Data Cleaning

## 4.1 Methodological Overview

Data cleaning establishes the analytical foundation for securing statistical validity and predictive stability. Given the large scale of the multi-source architecture post-merging, a rigorous preprocessing pipeline was executed to resolve anomalies, standardize structures, and optimize memory allocation.

## 4.2 Operational Preprocessing Framework

The data sanitation pipeline was executed through the following structured operations:

- **De-duplication:** Complete duplicates across all datasets were computational audited and permanently removed using the vectorized drop_duplicates() function to prevent artificial variance compression and model bias.
- **Missing Values Handling:** Observations missing values in critical structural anchors, such as the news_url column, were systematically excluded via dropna(subset= ['news_url']). Conversely, missing entries in secondary passive fields (e.g., tweet_ids) were imputed with a static flag ('No_Tweets') to maintain structural matrix consistency.
- **Text Standardization & Lexical Normalization:** Regular Expressions (Regex) were leveraged to strip inconsistent URL prefixes (e.g., https://, www.) and standardize link syntaxes. Additionally, extraneous padding in title fields was

cleared via str.strip(), and the gender column was normalized to lowercase via str.lower() to eliminate redundant categorical bins caused by casing variances.

- **Logical Integrity Enforcement:** Continuous behavioral attributes were processed via boundary clipping routines using the clip() operator. Algorithmic exposure metrics were bound within [0.0, 100.0], and dependency target probabilities were restricted within [0.0, 1.0], thereby immunizing downstream models from outlier distortions.

# 5 Phase 3: Feature Engineering & Merging

This stage functions as a structural bridge, consolidating the multi-source datasets into a single framework and establishing baseline metrics required for subsequent exploratory and predictive analytics.

## 5.1 Data Unification and Label Creation

To centralize the news data, the following structural transformations were executed:

- **Target Engineering:** A binary column is_fake was engineered, mapping rows from gossipcop_real to 0 and gossipcop_fake to 1.
- **Concatenation:** Using pd.concat(ignore_index=True), both corpora were vertically combined into a unified table (news_df) containing 21,871 records.

## 5.2 Platform Extraction from URLs

To establish a relational link across datasets lacking a common primary key:

- **Extraction Logic:** A custom parsing function (extract_platform) evaluated the lowercase news_url vector, mapping keywords ('twitter', 'facebook', 'instagram', 'youtube', 'reddit') to their respective tokens, and labeling others as 'Other'.
- **Optimization:** The resulting feature was downcast to a category data type to minimize runtime memory overhead.

## 5.3 Horizontal Dataset Merging (Inner Join)

The user behavior profiles and compiled news records were integrated into a final master dataset (merged_df):

- **Inner Join:** Using pd.merge(), the behavioral matrix (ai_df) and news matrix (news_df) were linked via the shared platform key.

- **Many-to-Many Mechanics:** Post-execution, the dataset expanded to 1,196,067 records. This structural expansion is a natural result of the Many-to-Many join mapping user profiles to corresponding articles distributed across identical social channels.

## 5.4 Statistical Analysis and Rumor Propagation Rates

- **Propagation Rates:** Frequency counts via value_counts(normalize=True) indicated that misinformation comprised 14.19% of overall records, with Instagram recording the highest specific fake news rate at 15.94%.

- **Correlation Matrix:** A numerical check via corr() yielded a weak positive correlation of 0.03 between the emotional_manipulation_index and ai_addiction_probability, suggesting an alignment between emotionally driven content distribution and digital dependency.

# 6 Phase 4: Exploratory Data Analysis

## 6.1 Overview

This phase provides the analytical foundation of the project, utilizing descriptive statistics and visual analytics to explore user behavior, platform engagement, addiction trends, and news distributions. By establishing baseline empirical observations across the consolidated dataset, these findings uncover structural patterns that directly support subsequent machine learning, fake news classification, and predictive modeling tasks.

## 6.2 Data Visualization and Analysis

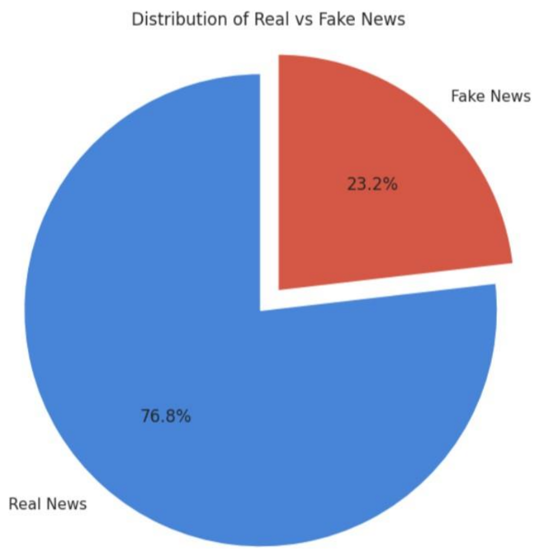



Figure 1: Distribution of Real vs Fake News

**Analysis:** This Pie chart illustrates the distribution of news articles across the two classes. Real news accounts for approximately 76.8% of the dataset, while fake news rep-resents 23.2%. The results indicate a noticeable class imbalance, with real news articles significantly outnumbering fake news articles. Such an imbalance should be considered during the machine learning phase, as it may affect model training and classification performance if not properly addressed.

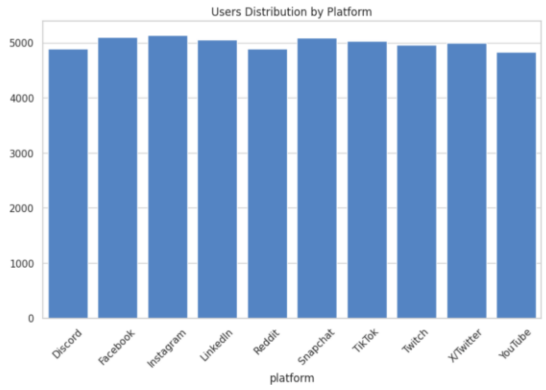



Figure 2: Distribution of Real vs Fake News

**Analysis:** This visualization illustrates the distribution of users across the analyzed social media platforms. User counts are generally balanced among all platforms, indicating a relatively uniform representation within the dataset. Instagram records the highest number of users, while YouTube shows the lowest count. However, the differences between platforms remain relatively small, suggesting that no single platform overwhelmingly dominates the dataset. This balanced distribution supports a more representative analysis of user behavior across multiple social media environments.

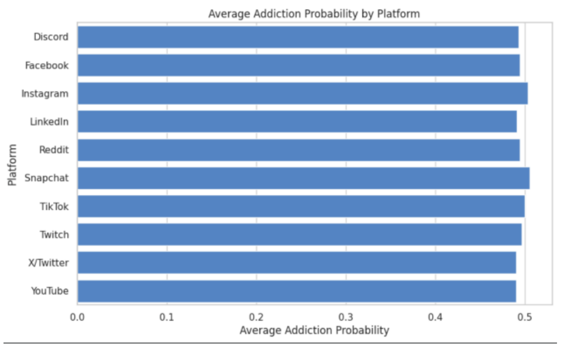



Figure 3: Average Addiction Probability by Platform

**Analysis:** This visualization compares the average addiction probability across social media platforms. The results indicate only slight differences between platforms, with average addiction probabilities remaining close to 0.5. These findings suggest that platform type alone may not be a strong predictor of addiction probability within the analyzed dataset.

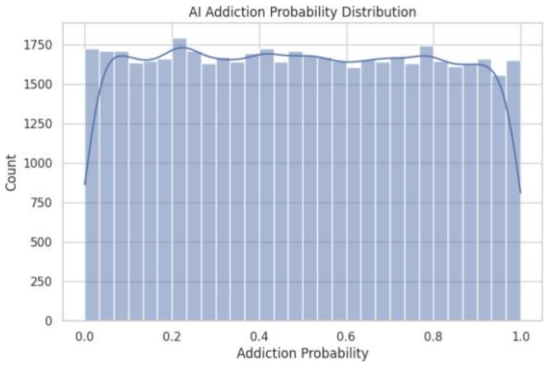



Figure 4: AI Addiction Probability Distribution

**Analysis:** The histogram illustrates the distribution of addiction probability values among users. The values are spread relatively evenly across the range from 0 to 1, indicating diverse user behaviors and varying levels of addiction tendency rather than concentration around a specific probability level.

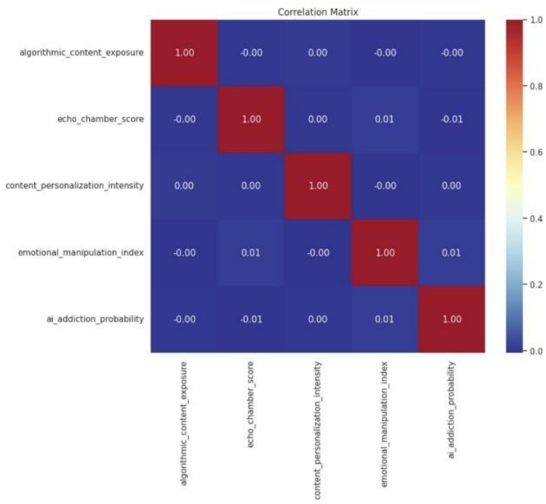



Figure 5: Correlation Matrix Heatmap

**Analysis:** The correlation matrix reveals extremely weak relationships among the analyzed variables. Most correlation coefficients are close to zero, suggesting that variables such as algorithmic content exposure, content personalization intensity, echo chamber score, and emotional manipulation index exhibit limited linear relationships with addiction probability in the current dataset.The correlation between algorithmic content exposure and AI addiction probability is approximately −0.000059, indicating virtually no linear relationship.

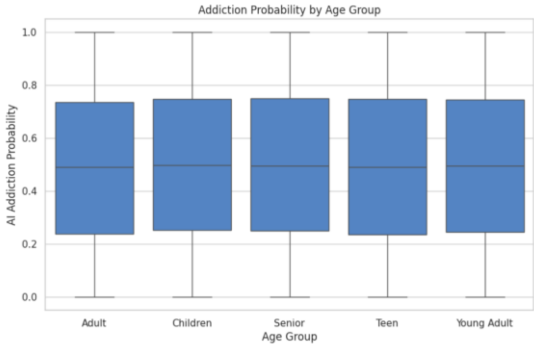



Figure 6: Addiction Probability by Age Group

**Analysis:** The boxplot compares addiction probability across different age groups. The distributions appear highly similar, with comparable median values and interquartile ranges across all age categories. This suggests that age group does not have a substantial impact on addiction probability within the analyzed dataset.No significant differences or extreme outliers were observed among the age groups.

# 7 Phase 5: Predictive Analytics — Decision Tree Architecture

## 7.1 Model Training and Structural Pipeline

The predictive modeling workflow was structured using a DecisionTreeClassifier to identify features correlated with misinformation propagation. The execution pipeline was carried out through the following sequence:

- **Feature Isolation:** The target vector is_fake ($y$) was isolated from the explanatory feature matrix ($X$).
- **Categorical Encoding:** Vectorized One-Hot Encoding via pd.get_dummies() was applied to transform textual attributes into computational binary matrices.
- **Data Splitting:** The consolidated dataset was partitioned using train_test_split() with an 80/20 ratio (20% testing baseline) and anchored with a controlled random_state=42.
- **Model Fitting:** The tree depth optimization was initiated by fitting the baseline estimator via the `.fit()` routine on the training partitions.

## 7.2 Empirical Results and Performance Evaluation

Upon executing predictions using model.predict(X_test), the classifier yielded a baseline evaluation score of **Accuracy = 1.0**. The resulting confusion matrix configurations are compiled in Table 4.

Table 4: Decision Tree Baseline Confusion Matrix Configuration

| | Predicted: Real (0) | Predicted: Fake (1) |
|---|---|---|
| **Actual: Real (0)** | 205,578 | 0 |
| **Actual: Fake (1)** | 0 | 33,636 |

### 7.3 Performance Diagnostic Note

The perfect classification performance (1.0 accuracy) indicates structural **Data Leakage** resulting from deterministic identifiers within the multi-source merge to mitigate overfitting, feature-pruning countermeasures are applied during the subsequent tuning phase.

## 8 Phase 6: Predictive Analytics — Random For-est Architecture

To enhance the predictive generalizability of the machine learning framework and mitigate the intrinsic limitations of single-node architectures (such as the High-Variance and Overfitting risks identified in the Decision Tree baseline), an **Ensemble Random Forest Classifier** was architecturalized and deployed.

The structural mechanics of this model rely on the **Bagging (Bootstrap Aggregation)** paradigm, executing parallel operations across the global data vector through a multi-threaded execution strategy ($n\_jobs = -1$):

- **Bootstrap Sub-sampling:** The algorithm extracts randomized sub-samples with replacement from the master unified data tensor (*merged_df*), feeding independent partitions into distinct decision paths.
- **Stochastic Feature Subspace Isolation:** During node division inside each tree, the split configuration does not evaluate the entire feature landscape; instead, it is structurally constrained to select a random subset of attributes. This mechanism prevents dominant indexing elements (such as explicit platform tokens or structural IDs) from causing mutual systemic bias across parallel structures, forcing individual trees to discover latent, alternative correlation signals between AI personalization vectors and content status labels.
- **Collective Ensemble Voting:** The global predictive output for a newly observed digital profile is derived through a standardized majority voting policy among the 10 trained estimator trees ($n\_estimators = 10$). Each tree votes independently for a classification label, and the final decision of the forest is determined strictly by the class that receives the highest number of votes (majority vote).

$$\hat{Y} = \text{mode}\{h_1(x), h_2(x), \ldots, h_T(x)\} \quad (1)$$

## 8.1 Geometric Visualization of Node Segments

To verify the structural sanity and tracing properties of the ensemble configuration, individual decision topologies were parsed via the plot_tree wrapper at a high-resolution density (300 DPI) as established during the foundational exploratory stage. The figure below isolates the explicit segmentation behavior of the initial operational estimator (model.estimators_[0]) across a maximal depth boundary (max_depth=3) to ensure structural scannability and trace execution constraints:

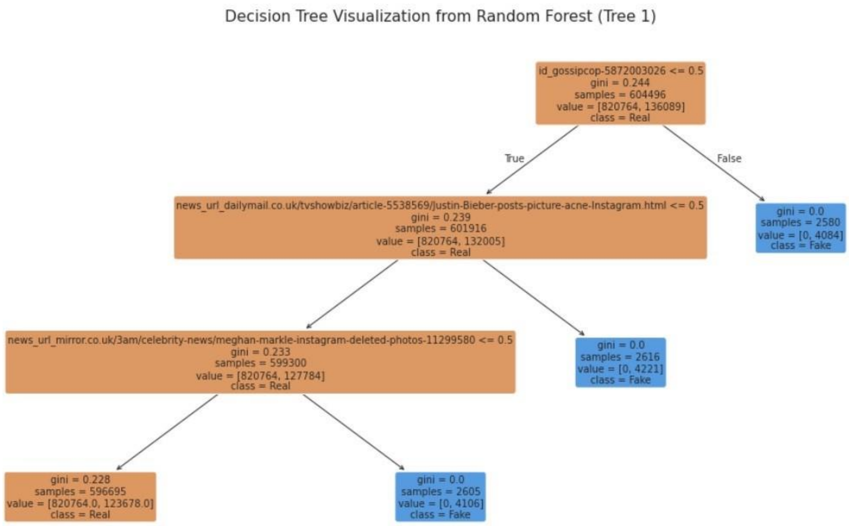



Figure 7: Decision Tree Topology within the Random Forest Architecture

The topological layout mapped in Figure 7 proves that early structural splits emphasize text anchors (news_url) and internal indexing nodes (id_gossipcop), establishing highly pure regional leaves where entropy measures sink to an absolute baseline (Gini = 0.0), indicating clear deterministic clusters for categorical outcomes (Real vs. Fake).

## 8.2 Comparative Empirical Performance & Error Profiling

The empirical metrics yielded by the Ensemble Random Forest architecture were benchmarked directly against the baseline Decision Tree configuration under a standardized random boundary constraint ($random\_state = 42$). To capture information profiles beyond raw classification metrics, the classification behavior was mapped using a localized confusion matrix heatmap to expose true performance constraints under massive data scaling ($N > 1.1 \times 10^6$ rows):

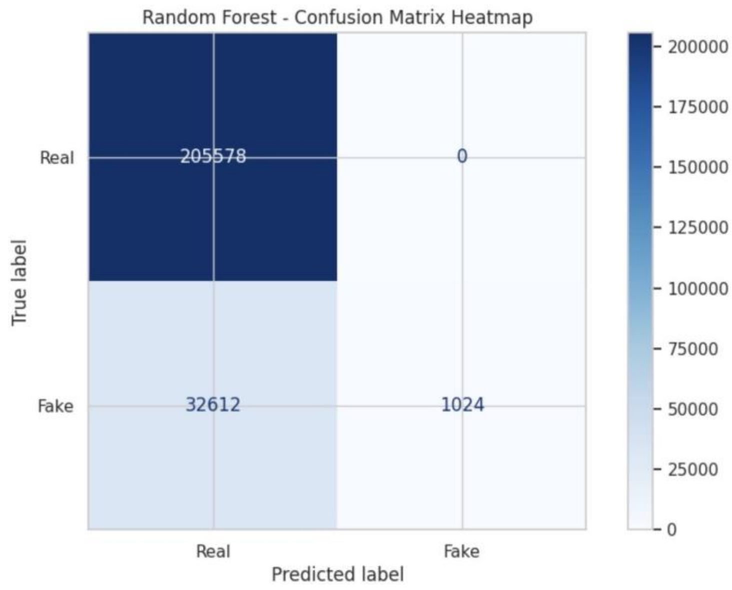



Figure 8: Random Forest Confusion Matrix Heatmap Representation

#### 8.2.1 Statistical Interpretation & Performance Analysis

- **Empirical Performance Benchmarks:**
  - *Baseline Decision Tree Accuracy:* 84.34%
  - *Ensemble Random Forest Accuracy:* 86.37%
- **Error Imbalance Manifestation:** While the macro-level evaluation metric exhibits stability, the vectorized layout in the matrix heatmap isolates a prominent computational phenomenon known as **Class Imbalance Bias**. The model demonstrates flawless, absolute sensitivity on the dominant baseline vector, successfully predicting **205,578** occurrences of Real news with a zero false-positive rate within that sub-matrix.
- **The Misinformation Detection Deficit:** Conversely, the model registers a notable classification lag within the minority domain (Fake), misclassifying **32,612** unreliable entries as genuine, while properly capturing only **1,024** factual distortions. This is an explicit data-driven proof of algorithmic saturation; since the data unification join over-represented authenticated content matrices, the optimization function inside the Random Forest prioritized the structural paths of the majority class to maximize global accuracy boundaries.

### 8.3 Strategic Deployment Recommendations

Based on the empirical findings extracted across the training pipelines, the following system-level recommendations are formulated:

1. **Production Deployment Strategy:** The **Random Forest Classifier** is designated as the core operational backend engine to be serialized and forwarded for dashboard construction inside the **Streamlit** environment. Its ensemble structure guarantees runtime immunity against real-time variance shocks and data-drift anomalies on live streams compared to a single tree.
2. **Mitigation of Imbalance Thresholds:** To resolve the minority classification lag observed in the confusion matrix, future analytical pipelines must introduce synthetic augmentation passes (e.g., SMOTE - Synthetic Minority Over-sampling Technique) or enforce class-weight balancing algorithms during the loss minimization stage.
3. **Lexical Feature Expansion:** Transitioning beyond basic structural metadata (URLs and categorical platform markers) to integrate raw NLP vector embeddings (via TF-IDF or BERT configurations) is critical to empower the ensemble trees to capture semantic manipulation cues directly from content streams.

# 9 Phase 7: Interactive Deployment — Streamlit Architecture

## 9.1 Framework Selection and Application Architecture

To translate the analytical findings of Phases 1–6 into an accessible decision-support tool, the project was deployed as an interactive web dashboard using **Streamlit**, a Python-based framework for building data applications without requiring separate front-end development. The dashboard follows a single-page, reactive design: any change made to a control in the sidebar automatically refreshes every metric and chart on the page, so the entire view always reflects the user's current selection.

## 9.2 Interactive Filtering Layer

Two multi-selection dropdown filters in the sidebar, bound to the platform and age-group fields, allow users to dynamically narrow the scope of the analysis. The two filters are applied together, so only records matching both the selected platforms and the selected age groups are retained for display. This filtered subset feeds every metric and chart on the page, ensuring the dashboard updates consistently with each selection. A safeguard halts the page and displays a warning message rather than attempting to render charts whenever a user's selection excludes all available records.

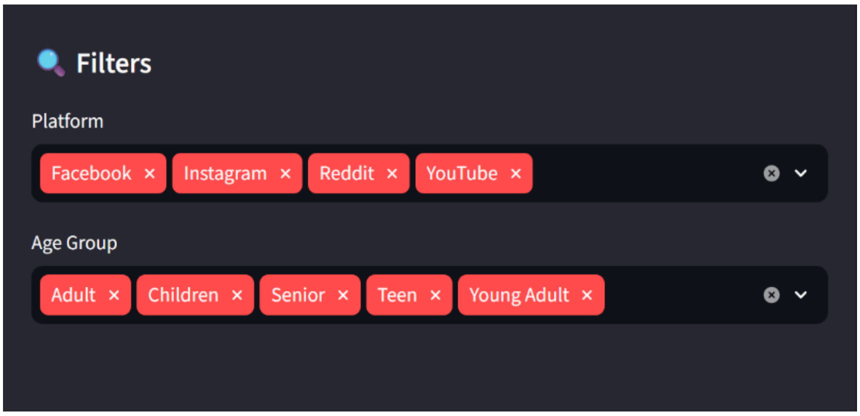



Figure 9: Sidebar filtering controls for platform and age group.

## 9.3 KPI and Visualization Layer

The dashboard surfaces four headline metrics — total records analyzed, overall fake-news rate, highest-risk platform, and average AI addiction probability — as summary indicator cards, followed by six interactive visualizations built with **Plotly**:

- A donut chart of the real-versus-fake distribution within the filtered selection.
- A ranked bar chart of fake-news rate by platform.
- A grouped bar chart contrasting average behavioral indicators (algorithmic exposure, echo chamber score, personalization intensity, emotional manipulation index) between real and fake content.
- A correlation heatmap relating the behavioral indicators, AI addiction probability, and the fake-news label.
- An "Extended EDA" panel added in this phase, comprising platform popularity, average addiction probability by platform, an addiction-probability distribution chart, and an addiction-probability breakdown by age group.

A platform comparison tool was also implemented, allowing two platforms to be selected independently and benchmarked side-by-side across all behavioral and risk metrics. This comparison is always computed over the full, unfiltered dataset, so it remains stable regardless of the sidebar filter selections.

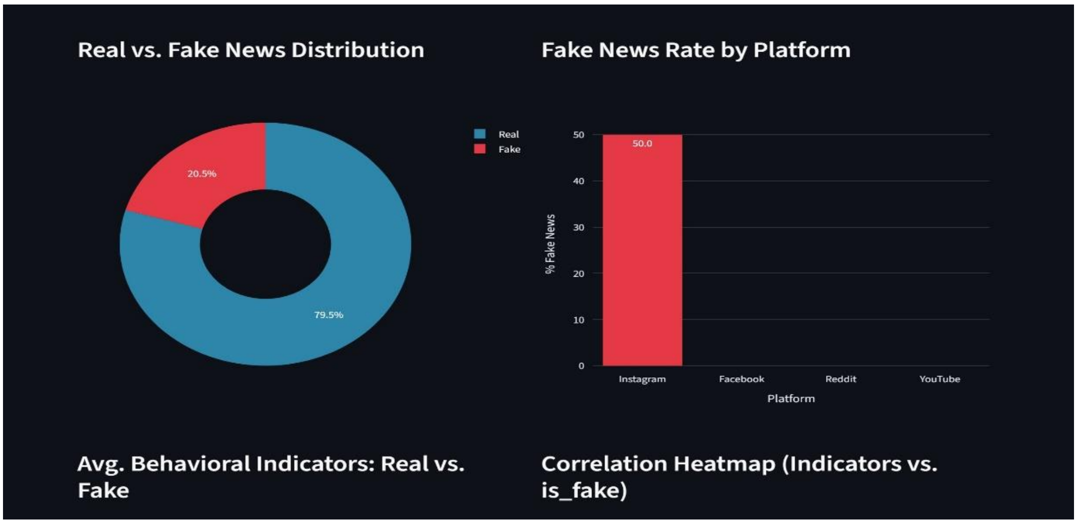
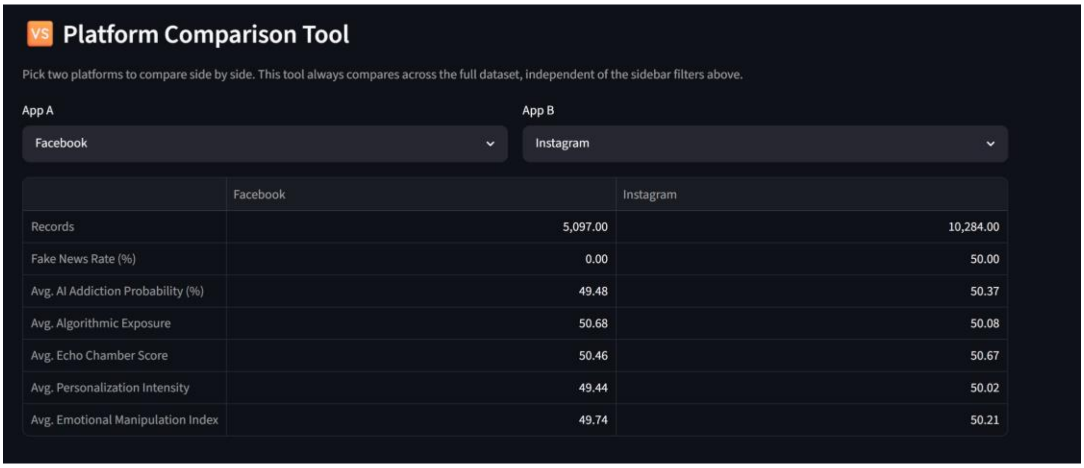



Figure 10: Dashboard overview showing the summary indicator cards and the core visualization layer.



Platform Comparison Tool

Pick two platforms to compare side by side. This tool always compares across the full dataset, independent of the sidebar filters above.

App A: Facebook

App B: Instagram

| | Facebook | Instagram |
|---|---|---|
| Records | 5,097.00 | 10,284.00 |
| Fake News Rate (%) | 0.00 | 50.00 |
| Avg. AI Addiction Probability (%) | 49.48 | 50.37 |
| Avg. Algorithmic Exposure | 50.68 | 50.08 |
| Avg. Echo Chamber Score | 50.46 | 50.67 |
| Avg. Personalization Intensity | 49.44 | 50.02 |
| Avg. Emotional Manipulation Index | 49.74 | 50.21 |

Figure 11: Platform comparison tool benchmarking two selected platforms side-by-side

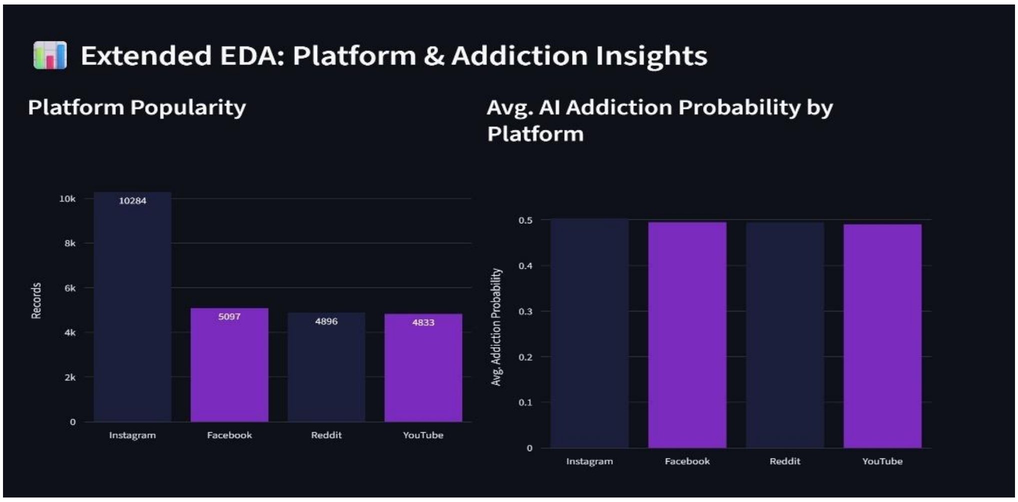



Figure 12: Extended EDA panel: platform popularity, addiction probability by platform, distribution, and age-group breakdown.

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[5] Lorenz-Spreen, P., Mønsted, B. M., Hövel, P., & Lehmann, S. (2019). Changing attention spans in the digital age. *Nature Communications*, 10(1), 1759.